# Governing AI by Evidence, Not Faith: A Cybernetic Architecture for Auditable Multi-Model Orchestration

証拠に基づく AI ガバナンス：検証可能なマルチモデル統制のためのサイバネティクス・アーキテクチャ

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## Abstract

Multi-model AI orchestration systems face a fundamental governance challenge: when multiple AI agents operate with partial autonomy across heterogeneous tasks, ensuring consistent, auditable, and correctable behavior requires institutional structure beyond prompt engineering or model-level constraints.

We present MoCKA (Model of Cybernetic Knowledge Architecture), a governance architecture grounded in three core mechanisms: (1) the Civilization Loop — an eight-stage mandatory governance cycle derived from industrial control engineering; (2) TRUST_SCORE — a quantitative inter-agent auditing mechanism that detects fabrication and inconsistency in AI outputs; and (3) DNA Packet Injection — a cross-session behavioral constraint protocol that restores governance-aligned behavior at session initialization.

Empirical evaluation across 255 sessions demonstrates: fault rate reduction of 62.7%, recurrence rate improvement from 0.17 to 1.00, autonomous execution risk (AER) reduction from 60% to 0%, and ZYXTS response quality improvement of +0.6367. The system additionally provides a fluid coordinate model P(t) = (X, Y, Z) for continuous governance health monitoring.

We report limitations honestly: single-operator deployment, 30-day observation window, and circular observability coefficient definition. The architecture is proposed as a replicable governance substrate for multi-model AI systems operating in institutional contexts.

*Keywords: AI governance, multi-agent systems, cybernetic architecture, observability, incident management, auditable AI, TRUST_SCORE, Civilization Loop*

# 1. Introduction

The deployment of AI systems has shifted from single-model assistants to multi-agent orchestration environments in which several models collaborate on extended tasks. This transition creates governance challenges that single-model safety approaches do not address.

The central problem is institutional: how can an organization trust outputs from AI systems it cannot fully verify, operating faster than human review cycles? Current frameworks provide normative guidance (NIST AI RMF [1], ISO/IEC 42001 [2]) but do not specify operational mechanisms for real-time multi-agent governance.

MoCKA addresses this gap through three operationalizable contributions: a mandatory eight-stage governance loop, a quantitative inter-agent trust scoring mechanism, and a cross-session behavioral constraint protocol. Together these form an institutional governance substrate — not a model-level constraint, but a system-level enforcement architecture.

**Design principle:** Structure over trust. Rather than relying on AI agents to behave correctly, MoCKA structurally enforces correct behavior through mandatory recording, cryptographic verification, and quantitative auditing.

## 1.1 Research Questions

RQ1: Does systematic event recording change AI agent behavior in multi-model orchestration?

RQ2: Can inter-agent trust be quantified to detect fabrication without human review of every exchange?

RQ3: Does cross-session behavioral constraint injection produce consistent governance-aligned behavior?

RQ4: Can institutional health be represented as a continuous trajectory rather than a binary state?

# 2. Related Work

## 2.1 AI Governance Frameworks

NIST AI RMF [1] provides a four-function framework for AI risk management. ISO/IEC 42001 [2] defines a management system standard. Both are normative — they specify what to do, not how to enforce it operationally. MoCKA complements these by providing operational mechanisms: mandatory event recording, cryptographic audit trails, and quantitative inter-agent trust scoring.

## 2.2 Multi-Agent Coordination

Research on multi-agent coordination [3, 4] treats coordination as an optimization problem. MoCKA treats it as a governance problem: the question is not how agents coordinate efficiently, but how their coordination can be made auditable and correctable.

## 2.3 Observability and Auditability

Systems observability [7] concerns metrics, logs, and traces for operational visibility. XAI research [5, 6] focuses on making model decisions interpretable. MoCKA adapts observability to AI governance: the target is institutional compliance, and the audit artifact is a cryptographically verified event ledger.

# 3. MoCKA System Architecture

## 3.1 Overview

MoCKA consists of four integrated components: the Civilization Loop (mandatory governance cycle), the Fluid Coordinate Model (continuous health monitoring), three core enforcement mechanisms, and a cryptographic audit layer. Figure 1 shows the full architecture.

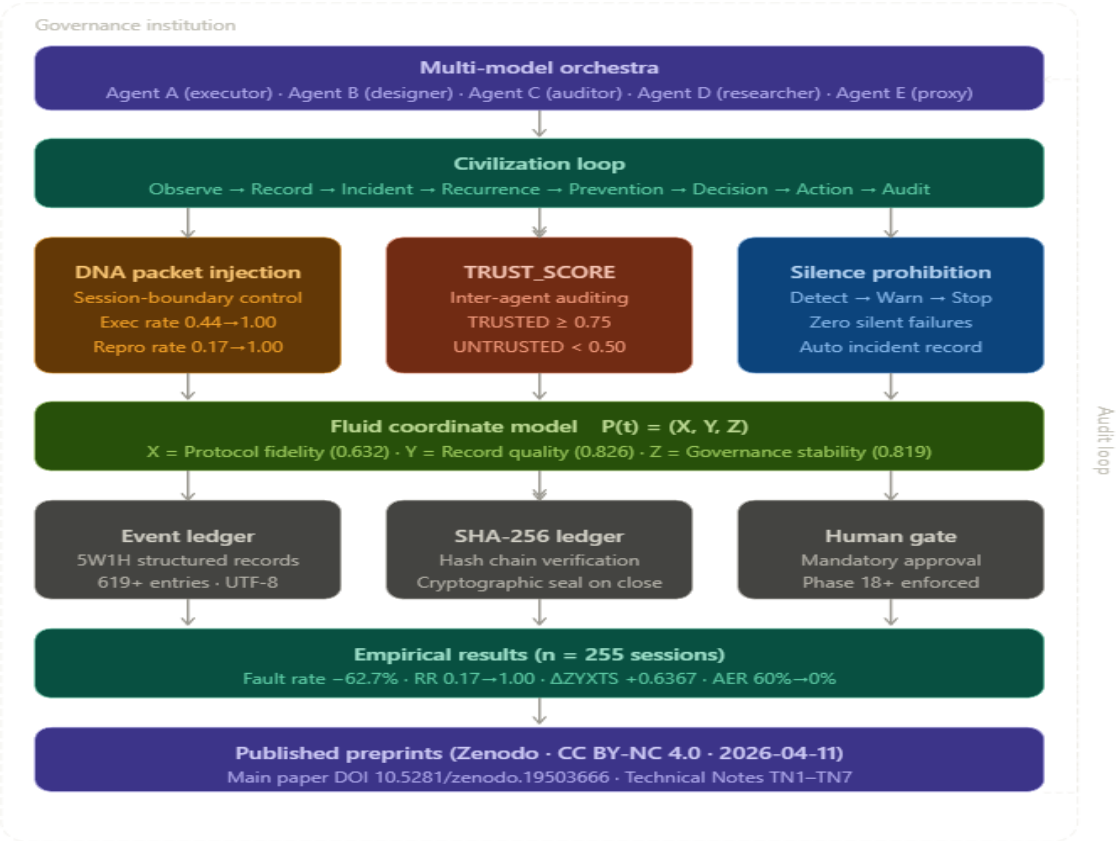

*Figure 1: MoCKA System Architecture — Governance institution with multi-model orchestra, civilization loop, three enforcement mechanisms, fluid coordinate model, and evidence layer.*

## 3.2 The Civilization Loop

The Civilization Loop enforces an eight-stage mandatory governance cycle for all institutional events (Figure 2). The design is derived from process safety management in industrial control systems, where stage-skipping is a safety violation.

*Figure 2: The Civilization Loop — Eight mandatory stages from Observation to Audit, with no stage skippable.*

| Stage | Operation | MoCKA Implementation | Skip consequence |
|---|---|---|---|
| ① Observe | Detect AI behavior | Observability coefficient α | Invisible AI action |
| ② Record | 5W1H structured log | validate_input_integrity() | Unverifiable history |
| ③ Incident | Classify risk level | risk_level ∈ {normal, warning, critical} | Risk invisibility |
| ④ | Pattern | recurrence_registry.c | Repeatable failure |

| | | | |
|---|---|---|---|
| Recurrence | analysis | sv | |
| ⑤ Prevention | Policy generation | policy_gen2.py | No institutional learning |
| ⑥ Decision | Human gate approval | Mandatory (Phase 18+) | Unauthorized change |
| ⑦ Action | Deploy change | Event-recorded deployment | Audit gap |
| ⑧ Audit | Cryptographic verify | SHA-256 hash chain seal | Unverified governance |

### 3.3 Fluid Coordinate Model P(t) = (X, Y, Z)

Governance health is represented as a continuous three-dimensional trajectory rather than a binary compliant/non-compliant judgment:

$$P(t) = (X(t), Y(t), Z(t))$$

| Axis | Name | Definition | Current value |
|---|---|---|---|
| X | Protocol fidelity | Fraction of interactions following institutional protocol | 0.632 |
| Y | Record quality | Mean completeness of 5W1H fields across events | 0.826 |
| Z | Governance stability | Consistency of enforcement over rolling 7-day window | 0.819 |

**Diagnostic technology effect:** Z-axis decline following MoCKA deployment (0.959→0.654) reflects improved detection of previously invisible failures, not actual degradation. Z subsequently recovered to 0.819 after corrective intervention.

## 4. Three Core Mechanisms

### 4.1 DNA Packet Injection

AI language models are stateless between sessions. DNA Packet Injection addresses this through compact JSON constraint packets injected at session initialization:

```
{"H": "MOCKA_DNA_v2", "G": 5.0, "C": "Strict", "A":
"attention_anchor", "P": "session_purpose"}
```

Packets are generated automatically by a four-layer pipeline: essence_classifier → essence_condenser → ping_generator → session initialization. The pipeline processes 230 operational essences condensed into 11 thematic batches.

| Metric | Pre-injection | Post-injection | Delta |
|---|---|---|---|
| Execution rate | 0.44 | 1.00 | +0.56 |
| Reproduction rate | 0.17 | 1.00 | +0.83 |
| INSTRUCTION_IGNORE incidents | 4 / 10 sessions | 0 / 10 sessions | −4 |

## 4.2 TRUST_SCORE: Quantitative Inter-Agent Auditing

TRUST_SCORE evaluates agent outputs across four dimensions:

| Dimension | Weight | Fabrication indicator |
|---|---|---|
| Structural integrity | 0.30 | Missing fields despite claimed completeness |
| Causal consistency | 0.25 | Effects without plausible causes |
| Temporal plausibility | 0.25 | Impossible sequences or future timestamps |
| Fabrication indicators | 0.20 | Fixed input values with variable statistical outputs |

Verdicts: TRUSTED (≥0.75), SUSPICIOUS (≥0.50), UNTRUSTED (<0.50). All evaluations are recorded in the event ledger with full 5W1H metadata.

**Motivating incident:** Agent B produced statistical outputs from a dataset in which all Z-axis values were fixed at 0.88 — a fabrication signature. Post-hoc TRUST_SCORE evaluation: 0.21 (UNTRUSTED).

## 4.3 Silence Prohibition Protocol

The Silence Prohibition Protocol mandates three invariant behaviors upon anomaly detection:

| Behavior | Requirement | Implementation |
|---|---|---|
| DETECT | Check input integrity before processing | validate_input_integrity() in write_safe_csv() |
| WARN | Output explicit warning — silence is prohibited | Print violation details to stdout + ledger |

| | | |
|---|---|---|
| STOP | Halt processing — no result produced | Return None; log incident automatically |

## 5. XYZ+ST Evaluation Framework

The XYZ+ST framework provides unified quantitative evaluation across two layers: XYZ tracks institutional governance state; ST measures individual AI response quality (Figure 3):

```
ZYXTS_total = (Z + Y + X + T + S) / 5
```

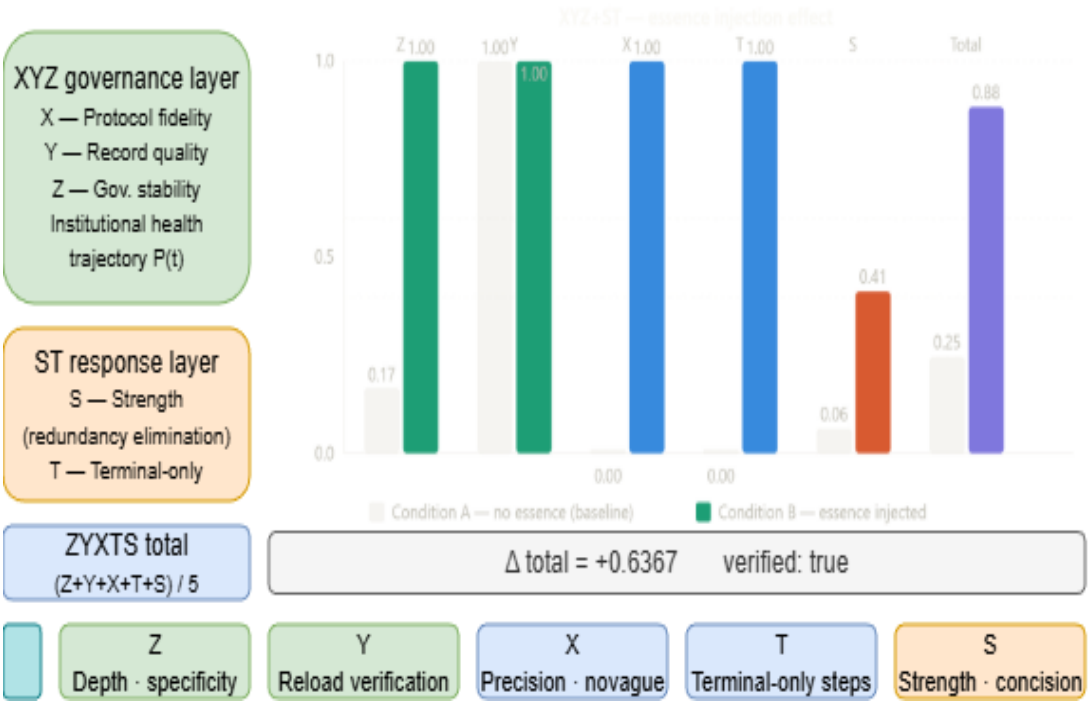


*Figure 3: XYZ+ST Evaluation — Condition A (baseline) vs Condition B (essence injected). ΔZYXTS = +0.6367, verified: true.*

| Axis | Layer | Name | Scoring criterion |
|---|---|---|---|
| Z | Governance | Depth/ specificity | Physical and technical vocabulary density |
| Y | Governance | Reload verification | Presence of re-verification language |
| X | Governance | Precision | Absence of vague/hedging language |
| T | Response | Terminal-only | Structured numbered steps to definitive conclusion |
| S | Response | Strength | Concision — information density relative to |

| | | | |
|---|---|---|---|
| | | | length |

### 5.1 Experimental Results

| Axis | Condition A (baseline) | Condition B (essence injected) | Delta |
|---|---|---|---|
| Z — depth | 0.1667 | 1.0000 | +0.8333 |
| Y — verification | 1.0000 | 1.0000 | 0.0000 |
| X — precision | 0.0000 | 1.0000 | +1.0000 |
| T — terminal | 0.0000 | 1.0000 | +1.0000 |
| S — strength | 0.0625 | 0.4125 | +0.3500 |
| TOTAL | 0.2458 | 0.8825 | +0.6367 |

ΔZYXTS = +0.6367, verified: true. X and T axes showed maximum improvement (+1.0 each), confirming that essence injection eliminates vague language and forces procedural completion.

## 6. Results

### 6.1 System-Level Metrics (n = 255 sessions)

| Metric | Pre-MoCKA | Post-MoCKA | Delta |
|---|---|---|---|
| Fault rate (λ) | 0.300 | 0.112 | −62.7% |
| Recurrence rate (RR) | 0.17 | 1.00 | +0.83 |
| Autonomous execution risk (AER) | 60% | 0% | −60pp |
| Observability coefficient (α) | 0.027 | ≈1.0 | +×36.96 |
| Record density (events/day) | 1.625 | 53.22 | +×32.7 |
| ZYXTS response quality | 0.2458 | 0.8825 | +0.6367 |

### 6.2 The Observability Paradox

MoCKA's primary function is establishing observability infrastructure, not suppressing incidents. Pre-MoCKA, α = 0.027 — only 2.7% of AI interactions were captured. Post-MoCKA, α ≈ 1.0. The governance chain:

```
MoCKA → α↑(×36.96) → Detection↑ → Response↑ → RR→0%
```

### 6.3 Temporal Risk Distribution

78% of high-risk events occurred between 00:00–04:00, empirically justifying the Human Gate design: autonomous AI operation during unmonitored hours generates disproportionate institutional risk.

## 7. Discussion

### 7.1 Structure Over Trust

MoCKA's central contribution is the operationalization of a governance principle: institutional structure is more reliable than agent trust. By making recording mandatory, verification cryptographic, and auditing quantitative, MoCKA removes the requirement that AI agents be trustworthy — instead requiring only that they be structurally constrained.

### 7.2 The A+B/2 Problem

AI agents exhibit a systematic behavioral pattern: when presented with conflicting requirements A and B, they produce outputs that partially satisfy both rather than making a definitive choice. DNA Packet Injection directly addresses A+B/2 through explicit priority ordering. TRUST_SCORE detects A+B/2 outputs through low causal consistency scores. The T-axis in ZYXTS scoring measures whether outputs reach a definitive terminal state.

### 7.3 Implications for Institutional AI Design

**Mandatory completeness:** Every governance stage must be completed. Optional stages become ignored stages.

**Cryptographic verification:** Governance records must be tamper-evident. SHA-256 hash chains provide this at negligible cost.

**Silence prohibition:** Agents that detect anomalies must report them. Silent failure is more dangerous than explicit failure.

**Quantitative trust:** Inter-agent trust must be measurable. TRUST_SCORE provides a replicable protocol.

## 8. Limitations

| Limitation | Description | Mitigation planned |
|---|---|---|
| Single operator | All 255 sessions conducted by one researcher in one environment | Multi-operator replication study |
| 30-day window | Observation period insufficient for | Extended longitudinal |

| | reliability engineering claims | study |
|---|---|---|
| Circular α definition | α_after = 1.0 is definitional, not measured | External audit of capture rate |
| Y-axis ceiling | ZYXTS Y-axis scored 1.0 in both conditions for test question used | Y-sensitive test case design |
| Model dependence | DNA injection relies on instruction-following ability | Testing across model types |
| Single-session ZYXTS | Conditions A/B evaluated within one session | Cross-session replication |

## 9. Conclusion

We presented MoCKA, a cybernetic governance architecture for multi-model AI orchestration. Through empirical analysis of 255 sessions, we demonstrated that systematic institutional structure produces measurable improvements in governance outcomes without requiring model-level modification.

Key quantitative contributions: fault rate reduction 62.7%, recurrence rate 0.17→1.00, AER 60%→0%, ΔZYXTS +0.6367. The fluid coordinate model P(t) = (X, Y, Z) provides continuous governance health monitoring that distinguishes genuine degradation from the diagnostic technology effect.

***"Governance structure controls reproducibility; institutional observability is the precondition for all other governance outcomes."***

MoCKA is available as open-source software. Technical Note series TN1–TN7 providing detailed specifications is published at DOI 10.5281/zenodo.19503666.

## References


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[7] Majors, C., Fong-Jones, L. & Miranda, G. (2022). Observability Engineering. O'Reilly.

[8] Wiener, N. (1948). Cybernetics. MIT Press.

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## Appendix: Technical Note Series

| Note | Title | Key contribution |
|---|---|---|
| TN1 | XYZ+ST Framework | Five-axis unified evaluation; ΔZYXTS=+0.6367 |
| TN2 | Silence Prohibition Protocol | DETECT→WARN→STOP; zero false negatives (5 categories) |
| TN3 | DNA Packet Injection | Cross-session constraint; execution rate 0.44→1.00 |
| TN4 | Fluid Coordinate Model | P(t)=(X,Y,Z); diagnostic technology effect |
| TN5 | TRUST_SCORE | Quantitative inter-agent auditing; TRUSTED/SUSPICIOUS/UNTRUSTED |
| TN6 | A+B/2 Problem | Averaging behavior definition; structural countermeasures |
| TN7 | Civilization Loop | Eight-stage mandatory cycle; industrial control origin |